

# Deep Augmented Performers

A New Ensemble Performance System by Fusion of Melody Morphing and Body Movements

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Figure 1: Automatic performance with Deep Augmented Performers

## ABSTRACT

The fusion of music-information processing and human physical functions will enable new musical experiences to be created. We developed an interactive music system called “Deep Augmented Performers,” which provides users with the experience of conducting a musical performance. This system converts music arranged using melody morphing into electrical muscle stimulation (EMS) to control the body movements of multiple performers. The melodies used in the system are divided into segments, and each segment has multiple variations of melodies. The user can interactively control the performers, thus the actual performance.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

## KEYWORDS

Melody morphing method; Electrical Muscle Stimulation (EMS)

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## 1 INTRODUCTION

We propose an interactive music system called “Deep Augmented Performers,” which enables melodies to be played by performers who are manipulated by a user. With Deep Augmented Performers, the user converts music arranged from melody morphing into EMS and controls the body movements of multiple performers, such as dancers and musicians. Variations of melodies are composed on the basis of the generative theory of tonal music (GTTM) [3], whereby switching the melody segments maintains the overall structure of the melody and only changes the ornamentation. We believe that a computer that can deeply understand music must be developed to achieve a composition support system that enables various musical operations. Therefore, we have been researching and developing systems based on GTTM such as those proposed by Lerdahl and Jackendoff, by introducing deep learning [2]. The time-span tree, which is a deep structure obtained as a result of a GTTM analysis, structurally shows the relationship between the important notes and ornamentation notes. One example of a melody manipulation using the time-span tree is melody morphing, which generates an intermediate melody from two input melodies.

## 2 MELODY MORPHING

Deep Augmented Performers enables melodies to be manipulated using melody morphing. Fig. 2 is an example of abstracting a melody by using a time-span tree. There is a time-span tree from melody D, which embodies the results of the GTTM analyses. The structurally important notes are connected to a branch nearer the root of the tree. In contrast, ornamentation notes are connected to the leaves of the tree. We can obtain an abstracted melody, melody E, by slicing the tree in the middle and omitting notes that are connected to branches under line E.

In melody morphing, we use the primitive operations of the subsumption relation (written as  $\sqsubseteq$ ), meet (written as  $\sqcap$ ), and join (written as  $\sqcup$ ). A subsumption represents the relation by which

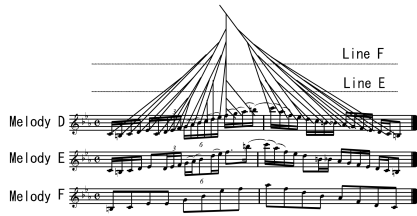


Figure 2: Time-span reduction of GTTM

“an instantiated object”  $\sqsubseteq$  “an abstract object.” The meet operator extracts the largest common part or the most common information of the time-span trees of two melodies in a top-down manner. The join operator joins two time-span trees in a top-down manner as long as the structures of two time-span trees are consistent.

By using time-span trees  $T_A$  and  $T_B$  from melodies A and B, we can calculate the most common pieces of information,  $T_A \sqcap T_B$ , which are the essential parts of melody A as well as those of melody B. The meet operations  $T_A \sqcap T_B$  are abstracted from  $T_A$  and  $T_B$ , and the discarded notes are regarded as the difference between  $T_A$  and  $T_B$ 's information (Fig. 3a). We argue that there are features without the other melody that are contained in the difference information of  $T_A$  and  $T_B$ . Therefore, we need a method for smoothly increasing or decreasing these features. The melody-divisional-reduction method can abstract the notes of the melody in the differential branch of the time-span tree (Fig. 3b). We use the join operator to combine melodies C and D, which are the results of the divisional reduction, using the time-span tree of melodies A and B (Fig. 3c).

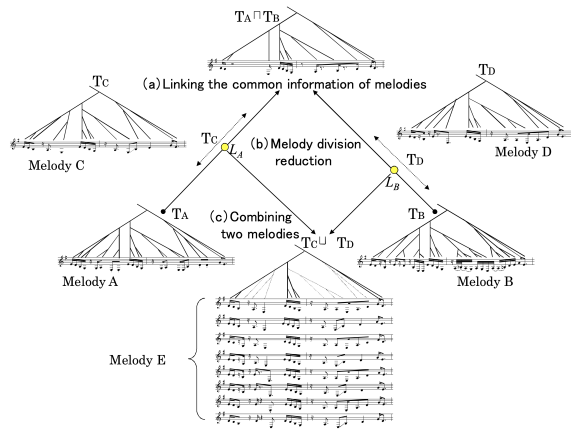


Figure 3: Melody morphing

### 3 DEEP AUGMENTED PERFORMERS

Deep Augmented Performers uses EMS to establish control over simple movements, such as snapping wrist joints and flexing elbow joints, of multiple people. The system includes a device that generates electrical stimulation by using an Arduino circuit (an electrical stimulation circuit), personal computer, direct current stabilized power supply, and electrode pads, with reference to a previous study [1] (Fig. 4). The stimulation pulse frequency is 40 to 70 Hz and the pulse width is 0.8 ms. The pulse voltage can be controlled by the user and varied depending type of muscle to be exercised. The system can synchronize the body movements of many people by controlling multiple EMS generators in an integrated manner. This makes it possible to play various types of ensembles (Fig. 4).

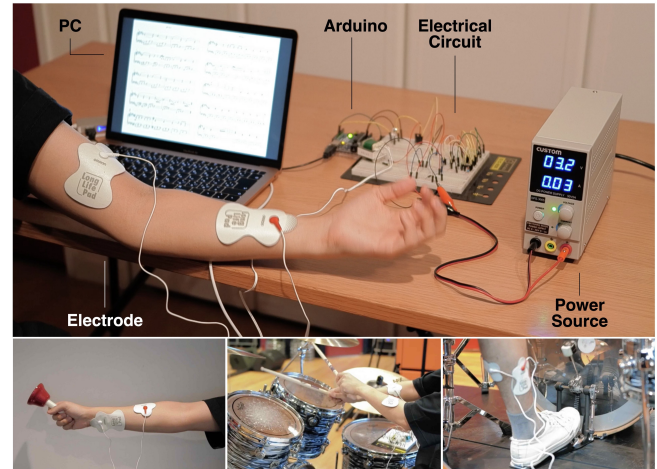


Figure 4: EMS system configuration and its feasibility study

The system is ideal for the automatic playing of ensemble percussion instruments. It can convert music arranged from melody morphing into EMS and control the body movements of multiple performers, via devices attached to the performers' hands and feet. The melodies used in the system are divided into multiple segments, and each segment has multiple melody variations. In addition, to enable anyone to easily manipulate melodies, we have developed a GUI that makes it possible to replace part of the melody. Thereby, the user can control performers by freely manipulating the melody.

We conducted a user survey on the effectiveness of the system that involved performances of four handbell players, one drummer, and one dancer (Fig. 1). We determined the correct electrode-pad position to reproduce the movements of each performer by considering the movement of the muscles. The handbell players could play the main melody. The drummer could play the corresponding four beats with full control by reading the tempo information of the song being played. For the dancer, we applied a weak electrical stimulus to his back by reading the tempo information of the song being played. The performance using this system lasted for about 5 minutes, and we confirmed that Deep Augmented Performers is effective by synchronizing the playing of the melody by using melody morphing and controlling performers' movements.

### 4 CONCLUSION

We proposed Deep Augmented Performers, which enables a user to manipulate music freely through melody morphing using the time-span tree of GTTM. The system controls the body movements of multiple performers to enable interactive performance. We will pursue a new style of musical experience by fusing music-information processing and human physical functions. We also plan to construct a system that enables more complex instrumental performances.

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